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Reg : FA 24- BSE-042

Class : BDS-4

Subject: Statics and Probability.

Assignment # 2

chapter #2 Probability.

Solutions-

=> 2.51-

Total envelopes = 500

Sample space for money = {100, 25, 10}

$$P[100] = \frac{75}{500}$$

$$P[25] = \frac{150}{500}$$

$$P[10] = \frac{275}{500}$$

Probability of envelop less than
100 dollars = $P[25] + P[10]$.

$$= P \frac{150}{500} + \frac{275}{500}$$

$$= \frac{150 + 275}{500}$$

$$= \frac{425}{500}$$

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 $\Rightarrow 2.53 -$

$$\text{Given: } P[S] = 0.7$$

$$P[B] = 0.4$$

$$P[SUB] = 0.8$$

$$(a) - P[S \cap B] = ?$$

by using formula

$$P[SUB] = P[S] + P[B] - P[S \cap B]$$

$$P[S \cap B] = P[S] + P[B] - P[SUB]$$

$$= 0.7 + 0.4 - 0.8$$

$$P[S \cap B] = 0.3$$

$$(b) - P[\bar{S} \cap \bar{B}] = 1 - P[SUB]$$

$$= 1 - 0.8$$

$$= 0.2$$

 $\Rightarrow 2.54$

$$\text{Given: } P[T] = 0.6$$

$$P[M] = 0.3$$

$$P[T \cap M] = 0.15$$

$$(a) - P[T \cup M] = ?$$

$$P[T \cup M] = P[T] + P[M] - P[T \cap M]$$

$$= 0.6 + 0.3 - 0.15$$

$$= 0.75$$

$$(b) - P[\bar{T} \cap \bar{M}] = 1 - P[T \cup M]$$

$$= 1 - 0.75$$

$$= 0.25$$

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$\Rightarrow$ 2.59.

Given: Total ways = $\begin{bmatrix} 52 \\ 6 \end{bmatrix}$

(a) - Probability of exactly 3 aces
choose 3 aces = $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$

Remaining 3 cards = $52 - 4 = 48$
 $\begin{bmatrix} 48 \\ 3 \end{bmatrix}$

Total favourable outcomes = $\begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 48 \\ 3 \end{bmatrix}$

$$\text{Probability} = \frac{\begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 48 \\ 3 \end{bmatrix}}{\begin{bmatrix} 52 \\ 6 \end{bmatrix}}$$

(b) - Probability of 4 hearts and
2 clubs.

$$4 \text{ hearts} = \begin{bmatrix} 13 \\ 4 \end{bmatrix}$$

$$2 \text{ clubs} = \begin{bmatrix} 13 \\ 2 \end{bmatrix}$$

$$\text{Probability} = \frac{\begin{bmatrix} 13 \\ 4 \end{bmatrix} \begin{bmatrix} 13 \\ 2 \end{bmatrix}}{\begin{bmatrix} 52 \\ 6 \end{bmatrix}}$$

$\therefore$ There are total 52 cards.

13 hearts, 13 diamonds, 13 clubs,

13 spaces (ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King).

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 $\Rightarrow 2.60$

Given: Total 9 = 5 novels, 3 books,
1 dict.

$$\text{Possible selection} = \binom{9}{3} = 84$$

$$(a) - 1 \text{ dict must} = \binom{8}{2} = 28$$

$$\text{Probability} = \frac{28}{84} = \frac{1}{3}$$

$$(b) - \text{novels} \binom{5}{2}, \text{books} \binom{3}{1}$$

$$= 10, = 3$$

$$= \frac{10 \times 3}{84} = \frac{30}{84} = \frac{5}{14}$$

 $\Rightarrow 2.61$

$$\text{Given: } S_T = 100, S_m = 54$$

$$S_H = 69, S_m \cap S_H = 35$$

$$(a) - P[S_m \cup S_H] = P[S] + P[M] - P[S_m \cap S_H]$$

$$= \frac{54}{100} + \frac{69}{100} - \frac{35}{100}$$

$$= \frac{88}{100}$$

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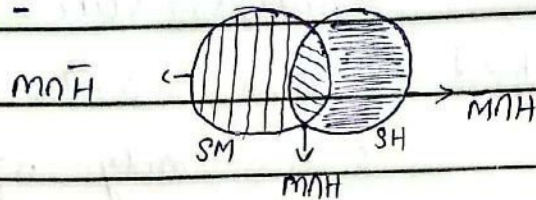
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$$(b) - P[\bar{S}_m \cap \bar{S}_u] = 1 - P[S_m \cup S_u]$$

$$= 1 - \frac{88}{100}$$

$$= \frac{12}{100}$$

(c) -



$$P[\bar{M} \cap H] = P[H] - P[M \cap H]$$

$$= \frac{69}{100} - \frac{35}{100}$$

$$= \frac{34}{100}$$

 $\Rightarrow 2.87.$

Given: Total key = 8

Correct = 1

Incorrect = 7

Total possible selection of 3 keys =

$$\begin{bmatrix} 8 \\ 3 \end{bmatrix} = 56$$

Total possible selection (wrong)

$$\text{key} = \begin{bmatrix} 7 \\ 3 \end{bmatrix} = 35$$

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$$\text{all wrong} = \frac{35}{56}$$

$$\text{collect} : 1 - \frac{35}{56} = \frac{21}{56} = \frac{3}{8}$$

$$\text{House unlocked} = 30\% = 0.30 \rightarrow$$

success sure. (1)

$$\text{House locked} = 70\% = 0.70 \rightarrow$$

success only if
collect key is picked.

$$\text{Combine probability} = 0.30 + \left(0.70 \times \frac{3}{8}\right)$$

$$= 0.30 + 0.2625$$

$$= 0.5625.$$

$\Rightarrow 2.89.$

$$\text{Given: } P[A] = 0.96 \quad P[B] = 0.96$$

$$P[A'] = 1 - P[A] \quad P[B'] = 1 - P[B]$$

$$= 1 - 0.96 = 1 - 0.96$$

$$= 0.04 = 0.04$$

$$(a) - P[A'AB'] = P[A'] P[B']$$

$$= (0.04)(0.04)$$

$$= 0.0016$$

$$(b) - P(A \cup B) = 1 - P[A'NB']$$

$$= 1 - 0.0016$$

$$= 0.9984.$$

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 $\Rightarrow 2.90$

A: The river is pulled.

B: A tested sample detects pollution.

C: Fishing is Permitted.

Given: $P(A) = 0.3$

$$P(B|A) = 0.75$$

$$P(B|A') = 0.20$$

$$P(C|A \cap B) = 0.20$$

$$P(C|A' \cap B) = 0.15$$

$$P(C|A \cap B') = 0.80$$

$$P(C|A' \cap B') = 0.90$$

$$\begin{aligned} (a) - P(A \cap B \cap C) &= P(A) \cdot P(B|A) \cdot P(C|A \cap B) \\ &= 0.3 \times 0.75 \times 0.25 \\ &= 0.045 \end{aligned}$$

$$(b) - P(B' \cap C) = \text{Two possibilities for } B' \{A \cap B', A' \cap B'\}$$

So,

$$\begin{aligned} P(B' \cap C) &= P(A \cap B' \cap C) + (A' \cap B' \cap C)P \\ &= 0.06 + 0.504 = 0.564 \end{aligned}$$

compute each

$$\begin{aligned} P(A \cap B' \cap C) &= P(A) \times P(B'|A) \times P(C|A \cap B') \\ &= 0.3 \times (1 - 0.75) \times 0.80 \\ &= 0.3 \times 0.25 \times 0.80 \\ &= 0.06. \end{aligned}$$

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$$\begin{aligned}
 P(A'NB'NC) &= 1 - P(A) \times 1 - P(B/A') \times P(C/A'NB') \\
 &= (1 - 0.3) \times (1 - 0.20) \times 0.90 \\
 &= 0.7 \times 0.80 \times 0.90 = 0.504
 \end{aligned}$$

(c) - $P(C) = \sum P(\text{each combo of } A, B, C)$

we already computed:

$$P(ANBNC) = 0.045$$

$$P(ANB'NC) = 0.06$$

$$P(A'NB'NC) = 0.504$$

we still need

$$P(A'NB'NC) = P(A') \times P(B/A') \times P(C/A'NB')$$

$$= 0.7 \times 0.20 \times 0.15$$

$$= 0.021$$

Now, sum all

$$= 0.045 + 0.06 + 0.504 + 0.021$$

$$= 0.63$$

(d) - $P(A/CNB')$

using Bay's rule. P

$$P(A/CNB') = \frac{P(ANB'NC)}{P(B'NC)}$$

$$= \frac{0.06}{0.564} = 0.1064$$

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 $\Rightarrow 2.99.$

Given:

$$\text{John} \rightarrow P[S_1] = 0.2 \rightarrow P(F/S_1) = 1/200 = 0.005$$

$$\text{Tom} \rightarrow P[S_2] = 0.6 \rightarrow P(F/S_2) = 1/100 = 0.01$$

$$\text{Jeff} \rightarrow P[S_3] = 0.15 \rightarrow P(F/S_3) = 1/90 = 0.011$$

$$\text{Pat} \rightarrow P[S_4] = 0.05 \rightarrow P(F/S_4) = 1/200 = 0.005$$

$$P[S_i/F] = \frac{P[S_i] \times P[F/S_i]}{P[F]} \quad \text{--- (i)}$$

$$\begin{aligned} \therefore P[F] &= P[F/S_1]P[S_1] + P[F/S_2]P[S_2] + \\ &\quad P[F/S_3]P[S_3] + P[F/S_4]P[S_4] \\ &= (0.005)(0.2) + (0.01)(0.6) + \\ &\quad (0.011)(0.15) + (0.005)(0.05) \\ &= 0.0089 \text{ put in (i)} \\ &= \frac{0.2 \times 0.005}{0.0089} \\ &= 0.1124. \end{aligned}$$

 $\Rightarrow 2.101.$

$$\text{Given: } P[L] = 0.75, P[L'] = 1 - P[L] = 0.25$$

$$P[R/L] = 0.6, P[R/S] = 0.3$$

$$P[L/R] = ?$$

using Bay's theorem.

$$P[L/R] = \frac{P[R/L]P[L]}{P[R]} \quad \text{--- (i)}$$

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$$\therefore P[R] = P[R|L] P[L] + P[R|S] P[S]$$

$$= 0.6 \times 0.75 + 0.3 \times 0.25 \rightarrow \text{put in ①}$$

1. becomes

$$= 0.6 \times 0.75$$

$$0.6 \times 0.75 + 0.3 \times 0.25$$

$$P(L|R) = 0.875$$

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Chapter #3

 $\Rightarrow$ 3.2

Given data:

3 Automobiles

 $B \rightarrow$ Blemish $N \rightarrow$ Non Blemish $X \rightarrow$ Random variable $\rightarrow$ No of Blemish

Truth Table

N N N 0

N N B 1

N B N 1

N B B 2

B N N 1

B N B 2

B B N 2

B B B 3

 $\Rightarrow$ 3.5Given: (a) - $f(x) = c(x^2 + 4)$
for $x = 0, 1, 2, 3$

$$f(x) = c(x^2 + 4)$$

$$= \sum_{x=0}^3 c(x^2 + 4)$$

$$x = 0$$

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$$+ c(0^2+4) + c(1^2+4) + c(2^2+4) + c(3^2+4) = 1$$

$$+ 4c + 5c + 8c + 13c = 1$$

$$+ 30c = 1$$

$$\boxed{c = 1/30}$$

$$(b) - f(x) = \binom{2}{x} \binom{3}{3-x} \text{ for}$$

$$x = 0, 1, 2$$

$$f(x) = c \binom{2}{x} \binom{3}{3-x}$$

$$= \sum_{x=0}^2 c \binom{2}{x} \binom{3}{3-x}$$

$$c \left[\binom{2}{0} \binom{3}{3} + \binom{2}{1} \binom{3}{2} + \binom{2}{2} \binom{3}{1} \right] = 1$$

$$\therefore \binom{a}{b} = \frac{a!}{b!(a-b)!}$$

$$c \left[\frac{2!}{0!2!} \times \frac{3!}{3!0!} + \frac{2!}{1!1!} \times \frac{3!}{2!1!} \right]$$

$$+ \frac{2!}{2!0!} \times \frac{3!}{1!2!}]$$

$$= c [1 + 6 + 3] = 1$$

$$c [10] = 1$$

$$10c = 1$$

$$\boxed{c = 1/10}$$

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 $\Rightarrow 3.6.$

formula am gonna use.

$$P[a \leq x \leq b] = \int_a^b \frac{20,000}{(x+100)^3}$$

$$(a) \quad a=200 \quad b=\infty \quad (b) \quad a=80 \quad b=120$$

using formula

$$= \int_{200}^{\infty} \frac{20,000}{(x+100)^3} dx = \left. \frac{-10,000}{(x+100)^2} \right|_{200}^{\infty}$$

$$= 20,000 \int_{200}^{\infty} (x+100)^{-3} dx = -10,000 - \left(\frac{-10,000}{(120+100)^2} \right)$$

$$= \frac{20,000 \cdot (x+100)^{-2}}{-2} \Big|_{200}^{\infty} = -10,000 - \left(\frac{-10,000}{(220)^2} \right)$$

$$= \frac{10,000 \times (x+100)^{-2}}{-1} \Big|_{200}^{\infty} = -10,000 - \left(\frac{-10,000}{32,400} \right)$$

$$= \frac{-10,000}{(x+100)^2} \Big|_{200}^{\infty} = \frac{-100}{324} - \left[\frac{-100}{484} \right]$$

$$= -0 - \left[\frac{-10,000}{(300)^2} \right] = \frac{25}{81} + \frac{100}{484}$$

$$= \frac{1}{9} \cdot \left(\frac{100}{484} \right) = \frac{25}{81} + \frac{25}{121}$$

$$= 25$$

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 $\Rightarrow 3.8$ Given: $P[H] = \frac{2}{3}$ $P[T] = \frac{1}{3}$

Event	H	T	$w(H-T)$
H H H	3	0	3
H H T	2	1	1
H T H	2	1	1
H T T	1	2	-1
T H H	2	1	1
T H T	1	2	-1
T T H	1	2	-1
T T T	0	3	-3

$$S = \{-3, -1, 1, 3\}$$

$$\begin{aligned}
 P(w) = -3 &= P\{[T T T]\} & P(w=3) &= P\{[H, H, H]\} \\
 &= P[T] P[T] P[T] & &= P[H] P[H] P[H] \\
 &= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} & &= \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \\
 &= \frac{1}{27} & &= \frac{8}{27}
 \end{aligned}$$

$$\begin{aligned}
 P(w=-1) &= P\{(HTT), (THT), (TTH)\} \\
 &= P[H] P[T] P[T] + P[T] P[H] P[T] + P[T] P[T] P[H] \\
 &= \left(\frac{2}{3} \times \frac{1}{3} \times \frac{1}{3}\right) + \left(\frac{1}{3} \times \frac{2}{3} \times \frac{1}{3}\right) + \left(\frac{1}{3} \times \frac{1}{3} \times \frac{2}{3}\right) \\
 &= \frac{2}{27} + \frac{2}{27} + \frac{2}{27} = \frac{6}{27} = \frac{2}{9}
 \end{aligned}$$

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$$\begin{aligned}
 P[w=1] &= P\{(HHT)(HTH)(THH)\} \\
 &= P[H]P[H]P[T] + P[H]P[T]P[H] + P[T]P[H]P[H] \\
 &= \left(\frac{2}{3} \times \frac{2}{3} \times \frac{1}{3}\right) + \left(\frac{2}{3} \times \frac{1}{3} \times \frac{2}{3}\right) + \left(\frac{2}{3} \times \frac{1}{3} \times \frac{2}{3}\right) \\
 &= \frac{4}{27} + \frac{4}{27} + \frac{4}{27} = \frac{12}{27} = \frac{4}{9}
 \end{aligned}$$

 $\Rightarrow 3.11$

Given:

7 $\rightarrow$ 2 defective7 $\rightarrow$ 3 random

$$\binom{2}{x} \binom{5}{3-x} \quad x = \{0, 1, 2\}$$

$$\binom{7}{3}$$

$$P[x=0]$$

$$\frac{2!}{3!} = \frac{\binom{2}{0} \binom{5}{3}}{\binom{7}{3}} = \frac{2}{7}$$

$$P[x=1] = \frac{\binom{2}{1} \binom{5}{2}}{\binom{7}{3}} = \frac{4}{7}$$

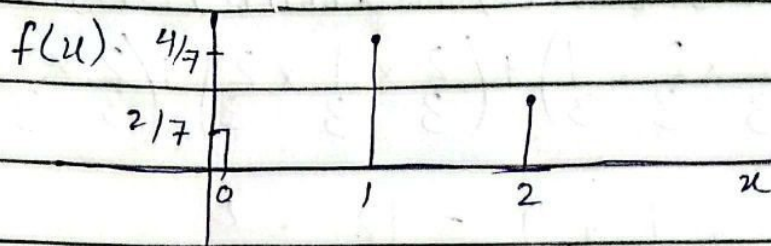
$$P[x=2] = \frac{\binom{2}{2} \binom{5}{1}}{\binom{7}{3}} = \frac{1}{7}$$

x	0	1	2
$f(x)$	$2/7$	$4/7$	$1/7$

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$$\frac{2}{7} + \frac{4}{7} + \frac{1}{7} = 1 \text{ PDF}$$



$\Rightarrow$ 3.14.

Given: x . 12 min

$$= \frac{12}{60} \text{ min}$$

$$= 0.2 \text{ hours}$$

(a) - CDF (Cumulative distribution function)

$$P[x < 0.2] = F[0.2]$$

$$= 1 - e^{-8 \times 0.2}$$

$$= 1 - e^{-1.6}$$

$$= 0.7981$$

(b) - PDF (Probability density function)

$$f(x) = F'(x) = \frac{d}{dx} F(x)$$

$$= \frac{d}{dx} (1 - e^{-8x})$$

$$= \frac{d}{dx} (1 - e^{-8x})$$

$$dx$$

$$= 8e^{-8x} = f(x)$$

$$= \int_0^{0.2} 8e^{-8x} dx$$

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$$= 8 \int_0^{0.2} e^{-8x} dx$$

$$= 8 \left. \frac{e^{-8x}}{-8} \right|_0^{0.2}$$

$$= -e^{-8x} \Big|_0^{0.2}$$

$$= -e^{-8 \times 0.2} - (-e)^{-8 \times 0}$$

$$= -e^{-1.6} + 1$$

$$= 1 - e^{-1.6} = 0.7981$$

=> 3.21.

Given: $f(x) = \int_0^1 k \sqrt{x} dx = 1$

$$1 = k \int_0^1 x^{1/2} dx$$

$$1 = k \left. \frac{x^{3/2}}{3/2} \right|_0^1$$

$$\frac{3}{2} = k \left. x^{3/2} \right|_0^1$$

$$\frac{3}{2} = k (1^{3/2} - 0^{3/2})$$

$\frac{3}{2}$	$= k$
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$$F(x) = \int_0^x k t^{1/2} dt$$

$$= \frac{3}{2} \int_0^x t^{1/2} dt$$

$$= \frac{3}{2} \times \frac{t^{3/2}}{3/2} \Big|_0^x$$

$$= x^{3/2} - 0^{3/2}$$

$$= x^{3/2}$$

$$F(x) = \begin{cases} 0, & 0 < x \\ x^{3/2}, & 0 \leq x < 1 \\ 1, & 1 \leq x \end{cases}$$

$$P[0.3 < x < 0.6] =$$

$$F(0.6) - F(0.3)$$

$$= (0.6)^{3/2} - (0.3)^{3/2}$$

$$= 0.3004$$

$\Rightarrow 3.27.$

Given: (a) - $f(x) = \frac{1}{2000} \exp(-x/2000) dt$

$$F(x) = \int_0^x \frac{1}{2000} \exp(-t/2000) dt$$

$$\frac{1}{2000} \int_0^x \exp(-t/2000) dt$$

$$\therefore \int a \exp(ax) dx = \exp(ax).$$

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$$F(x) = \exp(-x/2000) \Big|_0^x$$

$$= - \left[\exp\left(\frac{-x}{2000}\right) - \exp\left(\frac{0}{2000}\right) \right]$$

$$F(x) = \begin{cases} 1 - \exp\left(\frac{-x}{2000}\right), & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$(b) - P[x > 1000] = 1 - F[1000]$$

$$= 1 - \left[1 - \exp\left(\frac{-1000}{2000}\right) \right]$$

$$= 1 - 1 + \exp(0.5)$$

$$= \exp(-0.5)$$

$$= 0.6065$$

$$(c) - P[x < 2000] = F[2000]$$

$$= 1 - \exp\left(\frac{2000}{2000}\right)$$

$$= 1 - \exp(-1)$$

$$= 0.6321$$

$\Rightarrow 3.29.$

$$(a) - \int_1^{\infty} f(x) dx = \int_1^{\infty} 3x^{-4} dx$$

$$= 3 \int_1^{\infty} x^{-4} dx$$

$$= 3 \times \frac{x^{-3}}{-3} \Big|_1^{\infty}$$

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$$= -x^{-3} \Big|_1^{\infty}$$

$$= \left[\frac{1}{\infty^3} - \frac{1}{1^3} \right]$$

$$= -[0 - 1]$$

$$= 1$$

$$(b) - F(x) = \int_1^x 3t^{-4} dt = -t^{-3} \Big|_1^x$$

$$= - \left[x^{-3} - 1 \right]$$

$$= \begin{cases} 1 - x^{-3}, & x \geq 1 \\ 0, & x < 1 \end{cases}$$

$$(c) - P[x > 4] = 1 - F[4]$$

$$= 1 - (1 - 4^{-3})$$

$$= 1 - 1 + 4^{-3}$$

$$= 4^{-3}$$

$$= \frac{1}{4^3}$$

$$= \frac{1}{64}$$

$$64$$

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 $\Rightarrow 3.31.$

Given: $P(Y > 6) = F(6) - a$

$$f(y) \rightarrow F(y)$$

$$F(y) = \frac{1}{4} \int_0^y e^{-t/4} dt$$

$$= \frac{1}{4} \times \frac{e^{-t/4}}{-1/4} \Big|_0^y$$

$$= -e^{-t/4} \Big|_0^y$$

$$= -e^{-y/4} - (-e^{-0/4})$$

$$= -e^{-y/4} - (-1)$$

$$F(y) = 1 - e^{-y/4}$$

$$P(Y > 6) = 1 - F(6)$$

$$= 1 - [1 - e^{-6/4}]$$

$$= e^{-6/4} = 0.2231$$

(b) $P[Y \leq 1] = F(1)$

$$= 1 - e^{-1/4}$$

$$= 0.2212$$

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 $\Rightarrow 3.33.$

(a) - a valid pdf must satisfy

$$\int_0^1 f(y) dy = 1$$

$$\int_0^1 k y^2 (1-y)^3 dy = 1$$

$$k \int_0^1 y^2 (1-y)^3 dy = 1$$

$$\int_0^1 y^{a-1} (1-y)^{b-1} dy = \frac{F(a)F(b)}{F(a+b)}$$

$$\text{Here } y^2 = y^{3-1} \quad a=3$$

$$(1-y)^3 = (1-y)^{4-1} \quad b=4$$

$$\text{Thus } \int_0^1 y^2 (1-y)^3 dy = \frac{F(3)F(4)}{F(7)}$$

$$F(3) = 2! = 2$$

$$F(4) = 3! = 6$$

$$F(7) = 6! = 720$$

$$3(3,4) = \frac{2 \cdot 6}{720} = \frac{12}{720} = \frac{1}{60}$$

$$k \cdot \frac{1}{60} = 1$$

$$k = 60.$$

(b) - $P(Y \leq 0.5)$

$$\int_0^{0.5} 60 y^2 (1-y)^3 dy$$

$$(1-y)^3 = 1 - 3y + 3y^2 - y^3$$

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$$y^2 (1-y)^3 = y^2 - 3y^3 + 3y^4 - y^5$$

$$60y^2 - 180y^3 + 180y^4 - 60y^5$$

$$\int 60y^2 dy = 20y^3$$

$$\int -180y^3 dy = 45y^4$$

$$\int 180y^4 dy = 36y^5$$

$$\int -60y^5 dy = -10y^6$$

So, the cdf is

$$F(y) = 20y^3 - 45y^4 + 36y^5 - 10y^6$$

$$y = 0.5 \text{ put}$$

$$F(0.5) = 0.65625.$$

$$(C) - F(0.8) = 1 - P(Y < 0.8)$$

$$F(0.8) = 20(0.8)^3 - 48(0.8)^4$$

$$+ 36(0.8)^5 - 10(0.8)^6$$

$$= 0.98304$$

$$P(Y \geq 0.8) = 1 - 0.98304$$

$$= 0.01696$$

$\Rightarrow 3.35.$

$$\text{Given. } P(X > 8) = 1 - P[X \leq 8]$$

$$\therefore P[X \leq 8] + P[X > 8] = 1$$

$$= 1 - [P[X=0] + P[X=1] + P[X=2] + \dots + P[X=8]]$$

$$P[X > 8] = 1 - \sum_{x=0}^8 P[X=x]$$

$$= 1 - \sum_{x=0}^8 e^{-6} \frac{6^x}{x!}$$

$$= 1 - e^{-6} \left[\frac{6^0}{0!} + \frac{6^1}{1!} + \frac{6^2}{2!} + \dots + \frac{6^8}{8!} \right]$$

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$$= 0.1528$$

$$b- P[X=2] = \frac{e^{-6} 6^2}{2!} = 0.0446$$

⇒ 3.39.

Given: $0 \rightarrow \overset{x}{3}$, $A \rightarrow \overset{y}{2}$, $B \rightarrow 3] \rightarrow 4$

Solution:

$$\text{Total } \binom{8}{4} = \frac{8!}{4!4!} = 70$$

$$f(x, y) = \frac{\binom{3}{x} \binom{2}{y} \binom{3}{4-(x+y)}}{70}$$

$$x = 0, 1, 2, 3 \quad 1 \leq x+y = 4$$

$$y = 0, 1, 2$$

$$(b) - P(x+y \leq 2) = f(0,1) + f(0,2) + f(1,0) + f(2,0) + f(1,1)$$

$$f(0,1) = \frac{\binom{3}{0} \binom{2}{1} \binom{3}{3}}{70} = 2/70$$

$$f(1,0) = \frac{\binom{3}{1} \binom{2}{0} \binom{3}{3}}{70} = 3/70 \dots$$

$$\text{final answer} = 1/2.$$

⇒ 3.40

$$\text{Given: } f(x, y) = \frac{2}{3} (x+2y), 0 \leq x \leq 1,$$

$$0 \leq y \leq 1.$$

(a) Marginal density of x .

$$f_x(x) = \int_0^1 f(x, y) dy$$

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$$= \int_0^1 \frac{2}{3} (x+2y) dy$$

$$= \frac{2}{3} \left[\int_0^1 x dy + \int_0^1 2y dy \right]$$

$$= \frac{2}{3} (x+1).$$

(b)- Marginal density of y.

$$f_Y(y) = \int_0^1 f(x,y) dx$$

$$= \int_0^1 \frac{2}{3} (x+2y) dx$$

$$= \frac{2}{3} \left[\int_0^1 x dx + \int_0^1 2y dx \right]$$

$$= \frac{2}{3} \left(\frac{1}{2} + 2y \right)$$

$$= \frac{1}{3} + \frac{4}{3} y$$

(c)- $P(X < 1/2)$

$$= \int_0^{1/2} f_X(x) dx$$

$$= \int_0^{1/2} \frac{2}{3} (x+1) dx$$

$$= \frac{2}{3} \int_0^{1/2} (x+1) dx$$

$$= \int_0^{1/2} (x+1) dx = \frac{x^2}{2} + x$$

$$= \frac{(1/2)^2}{2} + \frac{1}{2} = \frac{1}{8} + \frac{1}{2} = \frac{5}{8} = \frac{2 \times 5}{3 \times 8}$$

$$= \frac{10}{24} = \frac{5}{12}$$

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 $\Rightarrow 3.42.$

$$P[X|Y=2] = \frac{P[X,Y]}{P[Y]}$$

$$P[Y] = \int f(x,y) dx$$

$$= \int_0^{\infty} e^{-(x+y)} dx$$

$$= \int_0^{\infty} e^{-x} e^{-y} dx$$

$$= e^{-y} \left[\frac{e^{-x}}{-1} \right]_0^{\infty}$$

$$= -e^{-y} [e^{-\infty} - e^0]$$

$$= e^{-y}$$

$$= e^{-y}$$

put in (1)

$$= \frac{e^{-(x+y)}}{e^{-y}}$$

$$= e^{-x}$$

$$= \frac{e^{-x} \cdot e^{-y}}{e^{-y}}$$

$$= e^{-x}$$

$$\int_0^1 e^{-x} dx = P[0 < x < 1 | Y=2]$$

$$= \left[-e^{-x} \right]_0^1 = -e^{-1} = 0.6321.$$

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→ 3.44.

Given:

$$f(x, y) = k(x^2 + y^2) \quad \begin{matrix} 30 \leq x < 50 \\ 30 \leq y < 50 \end{matrix}$$

a → Find k

$$= \int_{30}^{50} \int_{30}^{50} k(x^2 + y^2) dx dy = 1$$

$$= k \left[\int_{30}^{50} \int_{30}^{50} dx dy + \int_{30}^{50} \int_{30}^{50} y^2 dx dy \right]$$

$$= k \left[\frac{20 \cdot 98000}{3} + \frac{20 \cdot 980000}{3} \right]$$

$$= k \cdot \frac{40 \cdot 98000}{3}$$

$$= k \cdot \frac{3920000}{3}, \quad k = \frac{3}{3920000}$$

$$(b) P(30 \leq X \leq 40, 40 \leq Y < 50)$$

$$P = \int_{30}^{40} \int_{40}^{50} k(x^2 + y^2) dx dy$$

$$P = k \left[\frac{10 \cdot 37000}{3} + \frac{10 \cdot 61000}{3} \right]$$

$$= k \cdot 10 \cdot (98000)$$

$$P = \frac{3}{3920000} = \frac{980000}{3920000} = \frac{1}{4}$$

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(c)- Both are undefined.

$$P(X < 40, Y < 40)$$

$$= \int_{30}^{40} \int_{30}^{40} k(x^2 + y^2) dy dx$$

$$= \frac{37}{196}$$

 $\Rightarrow 3.49$

a).

x	1	2	3
$g(x)$	0.10	0.35	0.55

(b)-

y	1	3	5
$b(y)$	0.20	0.50	0.30

(c)- $P[Y=3 | X=2] = \frac{f(x=2, y=3)}{g(x=2)}$

$$= \frac{0.10}{0.35} = \frac{10}{35} = \frac{2}{7}$$

 $\Rightarrow 3.50$

(a).

x	2	4
$g(x)$	0.40	0.60

(b)-

y	1	3	5
$h(y)$	0.25	0.50	0.25

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 $\Rightarrow 3.57.$

$$f(x, y, z) = kxy^2z \quad 0 < x < 1, \quad 0 < y < 1, \\ 0 < z < 2.$$

$$(a) - \int_0^1 \int_0^1 \int_0^2 kxy^2z \, dz \, dy \, dx = 1$$

$$k \left(\int_0^1 x \, dx \right) \left(\int_0^1 y^2 \, dy \right) \left(\int_0^2 2 \, dz \right) = 1$$

$$= k \cdot \frac{1}{2} \cdot \frac{1}{3} \cdot 2$$

$$= k \cdot \frac{1}{3} = 1$$

$$= k = 3.$$

(b) -

$$P\left(x < \frac{1}{4} \mid x > \frac{1}{2} \mid 1 < z < 2\right)$$

$$P = \int_0^{1/4} \int_{1/2}^1 \int_1^2 3xy^2z \, dz \, dy \, dx$$

$$P = 3 \cdot \frac{1}{32} \cdot \frac{1}{24} \cdot \frac{3}{2} = \frac{21}{512}.$$

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 $\Rightarrow 3.60.$

$$f(x, y, z) = \frac{4}{9}xyz^2 \quad 0 < x, y < 1,$$

$$0 < z < 3.$$

(a) Joint Marginal of X, Y .

$$f(x)(y)(x, y) = \int_0^3 \frac{4}{9}xyz^2 dx.$$

$$\frac{4}{9}xy \int_0^3 x^2 dz = \frac{4}{9}xy \cdot 9 = 4xy$$

$$f(x, y)(x, y) \quad 0 < x, y < 1.$$

(b) Marginal density of z .

$$f_2(z) = \int_0^1 \int_0^1 \frac{4}{9}xyz^2 dx dy dz$$

$$= \frac{4}{9}z^2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{2^2}{9}$$

$$= f_2(z) = 0 < z < 3$$

$$(c) P = \int_{1/4}^{1/2} \int_{1/3}^1 \int_2^3 \frac{4}{9}xyz^2 dz dy dx$$

$$P = \frac{4}{9} \cdot \frac{3}{33} \cdot \frac{4}{9} \cdot \frac{19}{3}$$

$$P = \frac{19}{162}$$

$$P = \frac{19}{162}$$

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(d).

$$f_{z|x,y}(z|x,y) = \frac{f(x,y,z)}{F_{xy}(x,y)}$$

$$= \frac{4/9 \, xyz^2}{4xy} = \frac{z^2}{9}$$

$$P(0 < z < 2) = \int_0^2 \frac{z^2}{9} dz$$

$$= \frac{1}{9} \cdot \frac{8}{3} = \frac{8}{27}.$$

